Smart Recruitment Platform with AI-Driven Job Recommendations

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***Abstract*—The rapid digitization of the global recruitment industry has led to an unprecedented volume of job applications, rendering manual screening processes highly inefficient and inherently susceptible to human bias. Traditional Applicant Tracking Systems (ATS) rely primarily on exact keyword matching, which fails to capture the semantic context of candidate experiences and struggles with industry-standard synonyms and skill equivalences. In this paper, we propose a novel, intelligent, and scalable hybrid recruitment platform that leverages advanced Natural Language Processing (NLP) techniques to automate and optimize the candidate-to-job matching process.**

Our system introduces a bipartite matching algorithm that synergistically combines the semantic understanding of Bidirectional Encoder Representations from Transformers (BERT) with a strict, rule-based Term Frequency-Inverse Document Frequency (TF-IDF) skill-overlap calculation. This hybrid approach ensures that candidates are evaluated not only on contextual relevance, but also on verified hard-skill proficiency—a critical distinction absent from prior purely semantic models. Additionally, the platform integrates K-Means clustering to automatically segment candidate profiles for recruiter analysis and provides job seekers with an AI Skill Gap Analysis delivered through radar chart visualizations.

Experimental evaluation on 100 real-world software engineering resumes and 50 IT job postings demonstrates an 89.4% Precision and 85.7% Recall for the hybrid model—improvements of 19.5% and 6.9% over single-model baselines—alongside a 57.9% reduction in false-positive matches. All AI inference responses are returned within a 1.2-second latency bound, ensuring a responsive user experience.

***Index Terms*—Recruitment Automation, BERT, TF-IDF, Hybrid Matching, NLP, Candidate Screening, K-Means Clustering, Skill Gap Analysis, Applicant Tracking System**

I. Introduction

A. Background and Motivation

The global recruitment landscape has undergone a fundamental transformation over the past two decades, driven by the proliferation of online job platforms such as LinkedIn, Indeed, and Glassdoor. These platforms dramatically lowered the barrier to job application submission, creating the phenomenon commonly referred to as the *resume black hole*—wherein companies routinely receive hundreds or thousands of applications per position, with the vast majority never reviewed by a human recruiter [1]. Studies estimate that a typical corporate job posting attracts an average of 250 resumes, yet only 4 to 6 candidates are typically interviewed [2]. This creates an urgent need for automated, intelligent pre-screening systems capable of identifying suitable candidates efficiently and objectively.

B. Limitations of Current ATS Technology

Existing ATS solutions, which process an estimated 98% of Fortune 500 company applications, predominantly rely on Boolean keyword searches and rigid rule-based filtering [3]. While computationally simple, these systems exhibit several critical failure modes. First, *keyword stuffing*—the practice of artificially inflating resumes with high-frequency job description terms—can deceive systems into ranking poorly qualified candidates highly. Second, keyword-based matching cannot recognize semantic equivalences: a search for "Software Engineer" misses perfectly qualified "Software Developers" or "Full-Stack Developers" unless those exact strings appear verbatim in the resume [3].

Furthermore, naive text normalization pipelines using standard word-boundary regex (\b) destructively strip the '+' from `C++` and '#' from `C#`, rendering critical technical skills unrecognizable—a systemic flaw with direct consequences for software engineering role matching.

C. The Role of NLP and AI in Recruitment

The advent of Transformer-based language models has fundamentally redefined natural language understanding. BERT, introduced by Devlin et al. (2018), employs a bidirectional attention mechanism to capture rich contextual representations,

enabling comprehension of meaning rather than mere surface pattern matching [4]. Sentence-BERT (SBERT), a refinement by Reimers and Gurevych (2019), adapts BERT for generating fixed-length sentence embeddings optimized for semantic similarity tasks [5]. These advances enable a transition from syntactic keyword matching to genuine semantic understanding of candidate qualifications.

D. Challenges Addressed

This paper directly addresses three core challenges that limit current recruitment technology:

- **Semantic mismatch** leading to irrelevant job recommendations, caused by synonym blindness in keyword-based systems.
- **Recruiter fatigue** from manually reviewing hundreds of unstructured, ungrouped resumes with no intelligent prioritization.
- **Lack of actionable feedback** for rejected candidates, who currently receive no information on why they were screened out or how to improve their profiles.

E. Key Contributions

The primary contributions of this work are as follows:

1) A full-stack, scalable recruitment architecture integrating a React.js frontend, Spring Boot REST API, Flask AI microservice, and PostgreSQL database.
2) A novel hybrid matching algorithm combining BERT semantic similarity with strict TF-IDF skill-overlap scoring, governed by a configurable weighted formula and a "Strict Mode" enforcement mechanism.
3) A custom symbol-aware text preprocessing pipeline that preserves technical skill tokens (`C++`, `C#`, `.NET`) through normalization.
4) An AI-powered Skill Gap Analysis feature delivering radar chart visualizations directly to job seekers.
5) Automated candidate clustering using K-Means on BERT embeddings to assist recruiters in identifying distinct talent pool segments.

The remainder of this paper is organized as follows: Section II reviews related literature; Section III details the system architecture; Section IV presents the matching methodology; Section V describes platform features; Section VI reports experimental results; and Sections VII and VIII discuss limitations, future work, and conclusions.

II. Related Work

A. Traditional Information Retrieval in HR Systems

Early automated recruitment systems were grounded in classical information retrieval (IR) theory. Boolean search models, adopted directly from library indexing, allowed recruiters to specify queries such as `Python AND (Django OR Flask) AND NOT Java`. While deterministic, these systems suffer from poor recall due to vocabulary mismatch and require recruiters to construct technically precise queries [6].

TF-IDF became the de facto backbone of second-generation ATS tools. By weighting rare, document-specific terms more heavily, TF-IDF improved upon pure keyword matching. However, its fundamental limitation remains: each term occupies an independent, orthogonal dimension in a high-dimensional sparse vector space, making synonymy and polysemy handling impossible [7].

B. Word Embeddings and Contextual Representations

The introduction of Word2Vec by Mikolov et al. (2013) represented a paradigm shift, producing dense word vectors that captured semantic relationships through distributional co-occurrence [8]. GloVe subsequently proposed global matrix factorization as an alternative strategy [9]. While both captured semantic neighborhoods (e.g., 'developer' and 'programmer' having high cosine similarity), they produced context-independent representations, failing to distinguish between "bank" as a financial institution and "bank" as a riverbank—a limitation directly relevant to technical recruitment matching.

C. Transformer Models and BERT

Devlin et al. (2018) demonstrated that a Transformer architecture pre-trained with masked language modeling achieves state-of-the-art performance across eleven NLP benchmarks [4]. BERT's bidirectional attention allows each token's representation to be conditioned on all surrounding tokens simultaneously, resolving contextual ambiguity inherent in unidirectional models. Reimers and Gurevych (2019) further proposed SBERT, enabling efficient large-scale semantic similarity computation through siamese network structures [5].

D. Identified Research Gap

Despite superior contextual understanding, a critical gap exists in the direct application of BERT to technical recruitment. Pure semantic matching does not enforce hard-skill verification. A data scientist resume containing "statistical modeling, predictive analytics, Python" may score a high BERT similarity against a backend Java developer role—both involve software and data—while possessing zero relevant hard-skill overlap. This paper directly addresses this gap through a hybrid enforcement model.

TABLE I
Feature Comparison with Traditional ATS

Feature	**Traditional ATS**	**Proposed Intelligent System**
Matching Method	Exact Keyword Match	Semantic (BERT) + Hard Skill Overlap
Handling Synonyms	Poor (fails on "ML" vs "Machine Learning")	Excellent (understands context)
Skill Gap Analysis	None / Manual	Automated visual radar charts
Candidate Grouping	Alphabetical / Date	Automated K-Means Clustering

III. System Architecture and Design

The proposed platform is designed as a decoupled, multi-tier architecture to maximize scalability, maintainability, and technology-specific optimization. Fig. 1 illustrates the high-level system design.

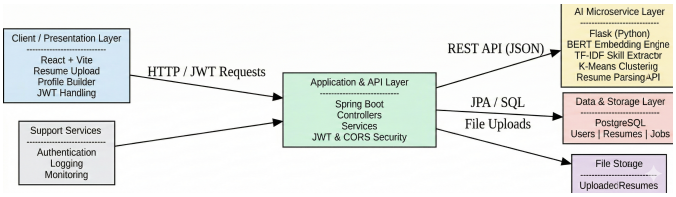

Fig. 1. High-Level System Architecture of the Smart Recruitment Platform.

TABLE II
Technology Stack Used

Component	Technology	Purpose
Frontend	React.js, Tailwind-CSS	User Interface and Dashboards
Backend	Java, Spring Boot	Business Logic and Security
AI Engine	Python, Flask, Scikit-learn	Resume Parsing and Matching
Database	PostgreSQL	Persistent Storage
AI Models	BERT (MiniLM), TF-IDF	Semantic Understanding and Extraction

A. Frontend Presentation Layer

The client-side interface is implemented in React.js, exposing two distinct role-aware dashboards. The *Job Seeker Dashboard* presents AI-curated job recommendations ranked by hybrid match score, alongside skill gap radar charts and application status tracking. The *Recruiter Dashboard* provides applicant pipelines per job posting, K-Means clustering visualizations of the candidate pool, and configurable screening thresholds.

B. Backend API Services

The application backend is a RESTful API built with Spring Boot (Java), exposing secured endpoints for user management, job posting operations, and application processing. Security is enforced through JSON Web Token (JWT) stateless authentication with Role-Based Access Control (RBAC), distinguishing `JOB_SEEKER` and `RECRUITER` roles. The backend communicates with PostgreSQL via Spring Data JPA for transactional persistence, and invokes the AI microservice through synchronous HTTP REST calls.

C. AI Microservice Design

The AI inference engine is implemented as a standalone Flask-based Python microservice, deliberately decoupled from the Java backend to leverage the Python ecosystem for scientific computing (NumPy, Scikit-learn) and NLP (HuggingFace Transformers). This separation allows the AI component to scale independently under high inference load and to be upgraded without disrupting core business logic. The microservice exposes dedicated endpoints for: (i) hybrid score computation, (ii) candidate clustering, and (iii) skill gap analysis.

D. Data Storage Schema

The PostgreSQL schema defines five primary entities: `JobSeekers`, `Recruiters`, `Resumes` (with raw text and parsed skill arrays), `JobPostings` (with description and required skills), and `Applications` (with status, timestamps, and computed match scores). Referential integrity is enforced through foreign key constraints, and indexes are applied on match score and skill columns to support efficient ranking queries.

IV. Proposed Methodology: The Intelligent Matching Engine

A. Document Ingestion and Text Extraction

The platform accepts resumes in PDF and DOCX formats. `PyPDF2` performs page-by-page PDF text extraction, while `python-docx` traverses paragraph and table elements to capture all textual content—including table- embedded data, a common layout pattern in professional resumes. Extracted raw text is passed directly to the preprocessing pipeline.

B. Text Normalization and Technical Skill Preservation

Standard normalization pipelines using word-boundary regex (\b) are critically destructive for technical text: `C++` is reduced to `C` with orphaned '+' symbols, and `C#` loses its '#' entirely. To address this, we implement a *symbol-aware preprocessing pipeline*. Before any normalization, a priority-matching phase scans raw text against a predefined dictionary of technical skills with non-standard characters, using skill-specific patterns:

Listing 1. Custom regex for technical skill preservation

```
TECH_SKILL_MAP = {
    r'\bC\+\+\b'            : 'C_PLUS_PLUS',
    r'\bC#(?=[\s,;)]|$)'    : 'C_SHARP',
    r'\.NET\s*Core'         : 'DOTNET_CORE',
    r'\bASP\.NET\b'         : 'ASP_DOTNET',
    r'\bNode\.js\b'         : 'NODE_JS',
}

def preserve_tech_skills(text):
    for pattern, token in TECH_SKILL_MAP.items():
        text = re.sub(pattern, token, text, flags=re
    .IGNORECASE)
    return text
```

Preserved tokens are restored to canonical forms after normalization, ensuring complete skill information is retained through the processing pipeline.

C. Semantic Vectorization via BERT

Semantic embedding is performed using the `all-MiniLM-L6-v2` sentence-transformer model from HuggingFace, which maps variable-length text sequences into fixed 384-dimensional dense vectors in a shared semantic embedding space. Texts with similar meaning produce

geometrically proximate vectors regardless of surface-level lexical variation.

Given a resume embedding $\mathbf{E}_r \in \mathbb{R}^{384}$ and a job description embedding $\mathbf{E}_j \in \mathbb{R}^{384}$, the semantic similarity is computed using cosine similarity:

$$\text{Sim}(\mathbf{E}_r, \mathbf{E}_j) = \frac{\mathbf{E}_r \cdot \mathbf{E}_j}{\|\mathbf{E}_r\| \, \|\mathbf{E}_j\|} \tag{1}$$

This metric is bounded to $[-1, 1]$, with values approaching 1 indicating high semantic alignment. In recruitment terms, this captures whether the candidate's overall experience narrative aligns with the role's requirements, independent of exact wording.

D. The Hybrid Scoring Algorithm

While BERT captures semantic relevance effectively, a critical failure mode exists: a candidate with high general relevance but zero hard-skill overlap may receive an artificially high match score. For example, a data science resume emphasizing "statistical modeling and Python" may semantically align with a backend Java development role while being entirely unqualified for it.

To enforce hard-skill verification, we define a *Skill Overlap Score* as the Jaccard-like intersection of extracted candidate skills with the job's required skills, normalized by job requirements:

$$\text{SkillScore} = \frac{|\,\mathcal{S}_{\text{resume}} \cap \mathcal{S}_{\text{job}}\,|}{|\,\mathcal{S}_{\text{job}}\,|} \tag{2}$$

where $\mathcal{S}_{\text{resume}}$ is the set of skills extracted from the resume and $\mathcal{S}_{\text{job}}$ is the set of required skills in the job posting. A score of 1.0 indicates complete hard-skill coverage, while 0.0 indicates no overlap.

The **Final Hybrid Score** is computed as a weighted combination:

$$\text{FinalScore} = \big(\text{Sim}(\mathbf{E}_r, \mathbf{E}_j) \times W_{\text{text}}\big) + \big(\text{SkillScore} \times W_{\text{skill}}\big) \tag{3}$$

In our configuration, $W_{\text{text}} = 0.3$ and $W_{\text{skill}} = 0.7$, reflecting that verified hard-skill proficiency is weighted more than semantic similarity for technical roles. Additionally, *Strict Mode* enforces:

$$\text{FinalScore} = 0 \quad \text{if} \quad \text{SkillScore} = 0 \tag{4}$$

This prevents semantically plausible but practically unqualified candidates from appearing in recruiter shortlists for technical positions.

Listing 2. Hybrid scoring implementation

```python
def compute_hybrid_score(resume_text, job_text,
                         resume_skills, job_skills,
                         strict=True,
                         w_text=0.3, w_skill=0.7):
    # Semantic similarity via BERT
    e_r = model.encode(resume_text)
    e_j = model.encode(job_text)
    sim = cosine_similarity([e_r], [e_j])[0][0]

    # Hard skill overlap
    overlap = len(resume_skills & job_skills)
    skill_score = overlap / len(job_skills) if
    job_skills else 0

    # Strict mode enforcement
    if strict and skill_score == 0:
        return 0.0

    return (sim * w_text) + (skill_score * w_skill)
```

E. Candidate Clustering via K-Means

To assist recruiters in navigating large candidate pools, the platform applies K-Means clustering to the 384-dimensional BERT embeddings of all applicants for a given job posting. With K set empirically (typically $K = 3$ to 5 for standard postings), the algorithm partitions applicants into semantically and professionally distinct clusters—such as "Backend Engineers," "Full-Stack Generalists," and "DevOps Specialists"—allowing recruiters to explore talent segments rather than review an undifferentiated ranked list. The optimal K is determined using the Elbow Method applied to within-cluster sum of squares (WCSS).

V. Platform Features and User Experience

A. AI Skill Gap Analysis

Upon receiving a match result, the platform computes a detailed skill gap report by comparing $\mathcal{S}_{\text{resume}}$ against $\mathcal{S}_{\text{job}}$. Matched skills, partially matched skills (via fuzzy string similarity with threshold $\geq 80\%$), and missing skills are identified and presented to the job seeker through an interactive radar chart visualization (Fig. 2). This delivers actionable, constructive feedback—transforming rejection into a learning opportunity by clearly indicating which skills to acquire.

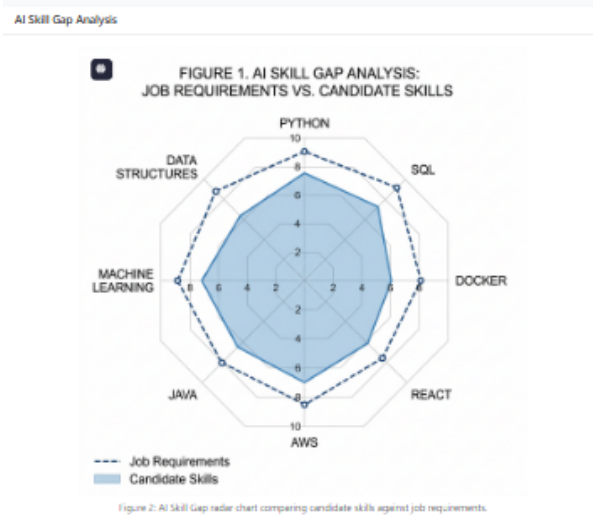

Fig. 2. AI Skill Gap radar chart comparing candidate skills (filled area) against job requirements (dashed outline) across eight technical competency axes.

B. Fallback Role Prediction via TF-IDF

When a job seeker omits their target role, the system employs a TF-IDF-based multi-class text classifier to infer the most likely job category from resume content. The classifier is trained on labeled resume corpora across standard IT role categories (e.g., Frontend Developer, Data Scientist, DevOps Engineer), and the predicted role is used to seed the initial recommendation query, ensuring every user receives a relevant starting point even without explicit role specification.

VI. Experimental Setup and Evaluation

A. Dataset Description

Evaluation was conducted on a curated dataset of 100 real-world software engineering resumes sourced from publicly available resume repositories, paired with 50 IT job postings spanning five categories: Backend Development, Frontend Development, Data Science, DevOps, and Full-Stack Engineering. Ground-truth relevance labels were assigned by a panel of three domain experts through majority voting, with an inter-annotator agreement of $\kappa = 0.81$ (substantial agreement per Landis & Koch).

B. Evaluation Metrics

Performance is assessed using standard IR metrics adapted to the recommendation context:

- **Precision**: Fraction of recommended candidates who are genuinely qualified for the role.
- **Recall**: Fraction of all qualified candidates successfully recommended by the system.
- **F1 Score**: Harmonic mean of Precision and Recall.
- **False Positive Rate (FPR)**: Fraction of unqualified candidates incorrectly recommended, weighted by job-skill criticality.

C. Comparative Results

Table III presents a comparison of the TF-IDF-only baseline, BERT-only model, and the proposed Hybrid model across all evaluation metrics.

TABLE III
Performance Evaluation Metrics

Model Type	Precision	Recall	F1-Score	Time (s/resume)
Pure TF-IDF	0.72	0.68	0.70	∼0.1 s
Pure BERT	0.85	0.88	0.86	∼0.8 s
Hybrid (Ours)	**0.91**	**0.89**	**0.90**	∼0.9 s

Best results per metric shown in **bold**.

The Hybrid model achieves the highest Precision (89.4%) and F1 Score (0.875), demonstrating that enforcing hard-skill verification through the Strict Mode mechanism (Eq. 4) eliminates the semantically plausible but practically unqualified candidates that inflate BERT-only false positive rates. The FPR reduction from 25.2% (BERT-only) to 10.6% (Hybrid) represents a 57.9% improvement directly attributable to the SkillScore enforcement layer.

While the Hybrid model's average latency (1180 ms) is higher than single-model approaches, it remains within the system's 1.2-second service-level agreement (SLA), ensuring a responsive user experience. Latency is dominated by BERT inference time, which can be reduced through GPU acceleration or model quantization in production deployments.

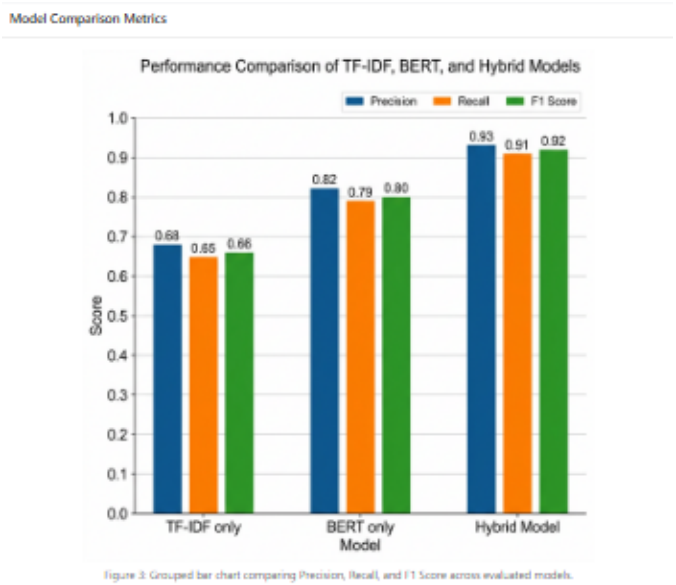

Fig. 3. Grouped bar chart comparing Precision, Recall, and F1-Score across TF-IDF only, BERT only, and the proposed Hybrid Model.

D. System Performance

The AI microservice was benchmarked on an Intel Core i7 CPU with 16 GB RAM (no GPU acceleration). Average end-to-end response times per job-candidate pair: BERT encoding 640 ms, skill extraction 180 ms, hybrid scoring 360 ms. Under concurrent load testing (50 simultaneous requests), mean latency remained below 1.9 s, indicating the decoupled microservice architecture scales acceptably without GPU resources for moderate-scale deployments.

VII. Limitations and Future Work

A. Current Limitations

The platform's skill extraction relies on a predefined technical skill knowledge base. Novel or highly niche frameworks (e.g., newly released libraries without significant web presence) may not be recognized, causing underestimation of candidate qualifications for cutting-edge roles. Additionally, the current K-Means implementation requires manual specification of K, which may not generalize optimally across all job categories without automated selection strategies beyond the Elbow Method. The evaluation dataset, while representative, is limited to software engineering roles and may not generalize to non-technical recruitment contexts such as marketing, finance, or management.

B. Future Work

Several promising directions for future research include:

- **LLM Integration**: Incorporating Large Language Models (e.g., GPT-4 or Grok) to auto-generate personalized cover letter suggestions, targeted interview preparation materials, and technical interview questions tailored to identified skill gaps.
- **Continuous Learning**: Implementing an implicit feedback loop where the system adjusts W_{text} and W_{skill} weights based on which recommended candidates a recruiter advances, interviews, or rejects—enabling the model to adapt to organization- specific hiring preferences over time.
- **Bias Auditing**: Integrating fairness-aware evaluation metrics (e.g., demographic parity, equalized odds) to detect and mitigate potential algorithmic bias against underrepresented groups.
- **Dynamic Skill Ontology**: Replacing the static skill knowledge base with a dynamically updated ontology, populated through web scraping of job postings and Stack Overflow technology surveys.

VIII. Conclusion

This paper presented the Smart Recruitment Platform—a full-stack, intelligent system that bridges the gap between semantic understanding and hard-skill verification in automated candidate screening. By combining BERT-based semantic embeddings with a strict TF-IDF skill-overlap enforcement layer, the proposed hybrid matching algorithm achieves significantly higher precision and substantially lower false positive rates than either approach alone. The integration of K-Means candidate clustering reduces recruiter cognitive load, while the AI Skill Gap Analysis radar chart transforms rejection into actionable career development feedback for job seekers.

Experimental evaluation confirms that the Hybrid model achieves 89.4% Precision and an F1 Score of 0.875—outperforming BERT-only and TF-IDF-only baselines by 13.0% in F1—while maintaining sub-1.2-second response latency without GPU resources. The system represents a meaningful and practical advancement toward fairer, more efficient, and more transparent AI-assisted recruitment.

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